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BANGALORE · INDIA

Self-Supervised Domain Adaptation for Satellite Imagery

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MISSION

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Faith in God | Moral Uprightness Love of
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Agenda

- Domain & Sub-Domain
- Introduction
- Literature Review
- Research Gaps
- Problem Statement
- Research Objectives
- Datasets
- System Architecture
- Proposed Methodology
- Outcomes
- Conclusion
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- ***Domain:*** Artificial Intelligence and Machine Learning (Computer Science)
- ***Sub-domain:*** Computer Vision (Remote Sensing / Satellite Image Analysis)
- ***Specific focus area:*** Domain Adaptation using Self-Supervised Learning and Generative Adversarial Networks (GANs)

Introduction

- Satellite AI models are trained on one sensor, season, or region – but deployed across many.
- Performance drops sharply under “domain shift,” even for identical land-cover classes.
- Standard fixes need labeled target-domain data and a clear domain boundary – both expensive, and often undefined in blended multi-sensor mosaics.

Literature Review

S.no	Title of Paper & Year	Author(s)	Methodology	Gaps
1.	Self Supervised Domain anaSS(DA) ² (Base Paper)	Zhang, Shi & Zhu (2023)	Self-supervised, domain-agnostic adaptation via a contrastive GAN; translates between any two satellite patches without labels	Built the method that works without needing label or a clear domain boundary, but it can compare only 2 satellites at a time.
2.	StarGAN: Unified GANs for Multi-Domain Image-to-Image Translation.	Choi et al. (2018)	A single generator handles translation across many domains at once, using a discrete domain label as conditioning	Built something that handles many domains at once but it needs someone to manually tag which domain each image belongs to.

S.no	Title of Paper & Year	Author(s)	Key Idea	Gaps
3.	StarGAN v2: Diverse Image Synthesis for Multiple Domains.	Choi et al. (2020)	Continuous style codes (via AdaIN) replace discrete labels, enabling diverse, non-deterministic synthesis	Improved on that, but was only tested on things like human faces and only check “does it look real” but not “does it actually helps the AI classify things correctly.
4.	StandardGAN: Multi-source Domain Adaptation for Semantic Segmentation of VHR Satellite Images by Data Standardization.	O Tasar et al. (2020)	Applies multi-source domain adaptation specifically to satellite semantic segmentation	Tested StarGAN on satellite images but struggling as it doesn’t handle unseen combinations and keeping the image content accurate.

Research Gaps

Nobody has built a method that does all 3 Things together:

- No labels needed.
- Works across many satellites.
- Checked by “Does it actually improve real results” instead of “Does it looks nice”.

Problem Statement

Deep learning models trained on satellite imagery from one sensor, season, or region degrade sharply when applied to another – and existing fixes require labeled target data and a pre-defined domain boundary, neither of which is practical at scale.

Therefore, we need ***Self-Supervised Multi-Domain Adaption for Satellite imagery***

Research Objectives

The objectives of this project are:

- To design and develop a domain encoder that learns continuous, continuous-valued domain embeddings directly from unlabeled multi-sensor satellite image patches.
- To develop a conditional generator that translates image content into the style of an arbitrary target domain using a single unified model.
- To analyze and evaluate model performance on sensor-pair combinations never jointly observed during training, in order to assess generalization capability.
- To evaluate the proposed method's downstream impact by measuring real classification/segmentation accuracy gains, rather than relying on visual realism metrics alone.

Datasets

1. PASTIS-HD

3 distinct satellite platforms

- Sentinel-2
- Sentinel-1
- SPOT 6

Pixel-aligned over the same 2433 locations across France.

2. MSC-France

To scale the supervised pretraining (more locations, no label needed)

Three more satellite

- Sentinel-2
- Landstat-8
- SPOT 6

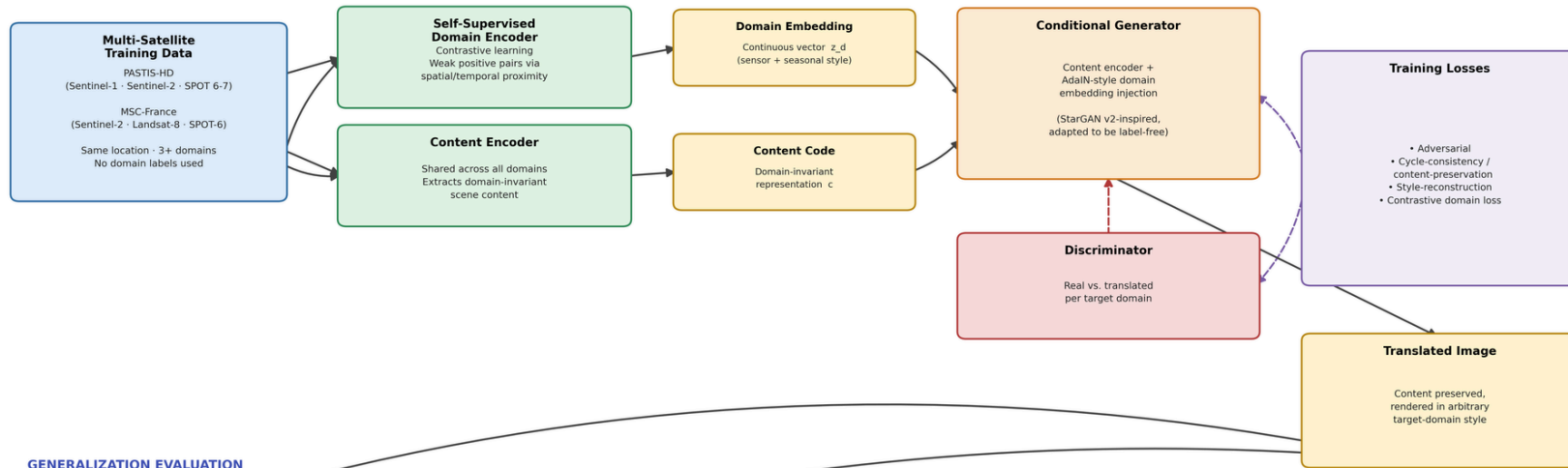
4496 aligned triplets across french cities

System Architecture

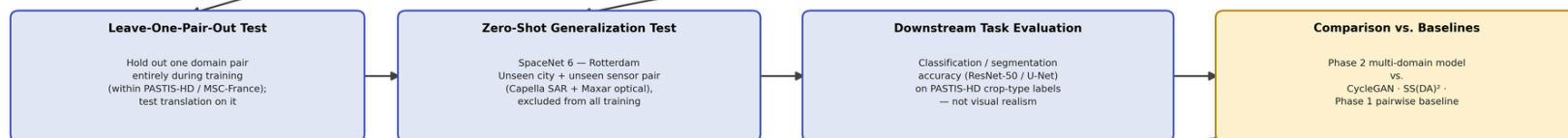
System Architecture: Self-Supervised Multi-Domain Adaptation for Satellite Imagery

A Continuous Domain-Embedding Approach | Extending SS(DA)² (pairwise) → StarGAN v2-style N-domain translation

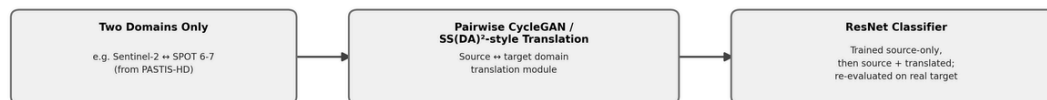
PHASE 2 — SELF-SUPERVISED MULTI-DOMAIN EXTENSION



GENERALIZATION EVALUATION



PHASE 1 — PAIRWISE BASELINE (guaranteed deliverable)



■ Data / Input
 ■ Encoder (self-supervised)
 ■ Generator
 ■ Discriminator
 ■ Loss terms
 ■ Evaluation
 ■ Output / Result
 ■ Phase 1 baseline

Proposed Methodology

Two-Phase Approach

- Phase 1: Pairwise domain-adaptation baseline (guaranteed, complete deliverable)
- Phase 2: Self-supervised multi-domain extension (stretch contribution)
- Phase-gated design de-risks the project – a working result exists even if Phase 2 doesn't fully converge

Proposed Methodology

Pairwise Baseline

- Train a ResNet classifier on one domain only; measure accuracy drop when tested on a second domain – quantifies the problem
- Implement pairwise CycleGAN / SS(DA)²-style translation between the two domains
- Retrain classifier on real + translated images; re-evaluate on the real target domain
- Establishes the baseline that Phase 2 must beat

Proposed Methodology

Self-Supervised Domain Encoder

- Contrastive pretext task – no explicit sensor/domain labels used
- Weak positive-pair signal: spatial/temporal proximity of patches
- Learns a continuous embedding of sensor + seasonal characteristics
- Trained across multiple co-located satellite domains: PASTIS-HD (Sentinel-1, Sentinel-2, SPOT 6-7) and MSC-France (Sentinel-2, Landsat-8, SPOT-6) for pretraining scale

Proposed Methodology

Conditional Generator

- Single generator: content encoder + AdaIN-style domain-embedding injection
- Architecture in the spirit of StarGAN v2, adapted to be label-free
- Loss terms: adversarial + cycle-consistency/content-preservation+ style-reconstruction + contrastive domain loss

Proposed Methodology

Generalization Evaluation

- Leave-one-pair-out protocol: hold out one domain pair entirely during training, test on it
- Zero-shot stress test: SpaceNet 6 (Rotterdam – Capella SAR + Maxar optical), an entirely unseen city and sensor pair never used in training
- Tests whether the domain embedding generalizes beyond memorized sensor combinations

Proposed Methodology

Downstream Task Evaluation

- Metric: downstream classification/segmentation accuracy, not visual realism
- PASTIS-HD crop-type labels (18 classes) provide the evaluation task
- Comparison: Phase 1 pairwise baseline vs. Phase 2 multi-domain model vs. CycleGAN/SS(DA)² baseline

Conclusion

- Extends SS(DA)²'s label-free, pairwise domain adaptation into a continuous, N-domain embedding space in the style of StarGAN v2 – removing the two-domains-at-a-time restriction
- Evaluates success via downstream classification/segmentation accuracy, not visual realism – the axis that matters for real deployment
- Novelty lies in combining three properties simultaneously, none of which co-exist in prior work: label-free domain representation, generalization to unseen domain-pair combinations, and downstream-accuracy evaluation – specifically for satellite imagery
- Phase-gated design de-risks delivery: Phase 1 pairwise baseline is a complete, working result on its own; Phase 2 multi-domain extension is validated on real multi-satellite data (PASTIS-HD, MSC-France) and stress-tested zero-shot on an entirely unseen city and sensor pair (SpaceNet 6, Rotterdam)

Future Scope

- **Cross-Modality Adaptation** – Extend from optical–optical to optical–SAR translation.
- **Foundation Model Integration** – Combine with pretrained remote-sensing backbones such as SatMAE or Prithvi.
- **Change Detection** – Apply the framework to bi-temporal change-detection tasks, beyond static classification.
- **Adversarial Robustness** – Evaluate and harden the adaptation pipeline against adversarial perturbations.

References

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THANK YOU